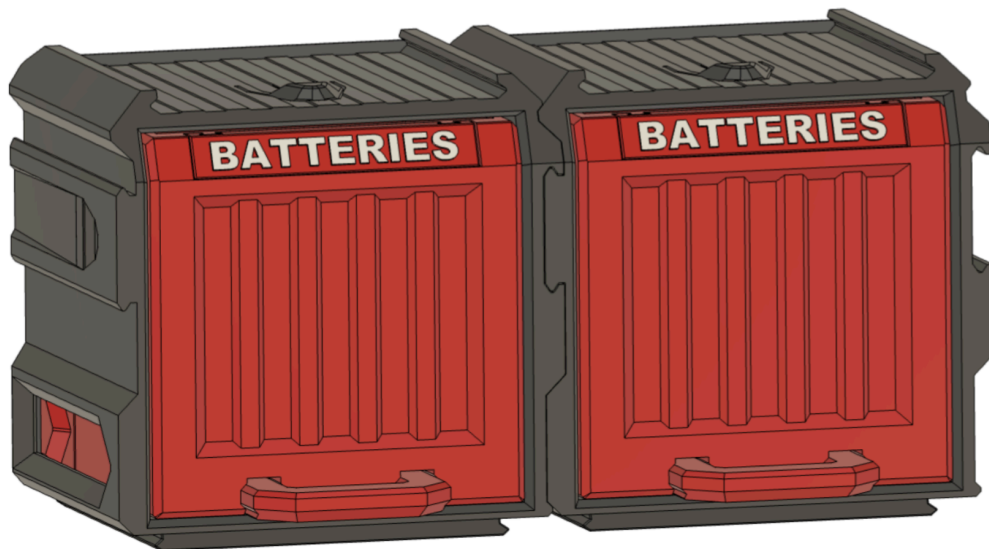




CARGO CONTAINERS: BATTERY STORAGE



User Guide

Resources, Print Settings and Assembly

Before You Start..

What's Included

The below files and documents are included within your download:

1. **Ready to Print (.3mf file)** – All the parts in one place and ready to print on Bambu Lab printers.
2. **User Guide (.pdf file)** – A complete 'User Guide' to help you print and assemble the model.
3. **Individual parts (.stl files)** – All the 3D printable parts you need to make the complete base model, compatible with most slicers.
4. **Assembly View Only file (Do Not Print)** – This file is to view the fully assembled model only and not for printing.

What We Used

Printers:

Bambu Lab P1S

You can get the printers we used here (*affiliate link**) – **[Bambu Lab P1S](#)**

Filaments:

Modular Case – Bambu Lab PLA Basic Dark Grey

Containers – Bambu Lab PLA Matte Scarlett Red and Marine Blue

You can get the filaments we used here (*affiliate link**) – **[Bambu Lab Filament](#)**

*Some links in this document are affiliate links, which means we may earn commission if you purchase through them.

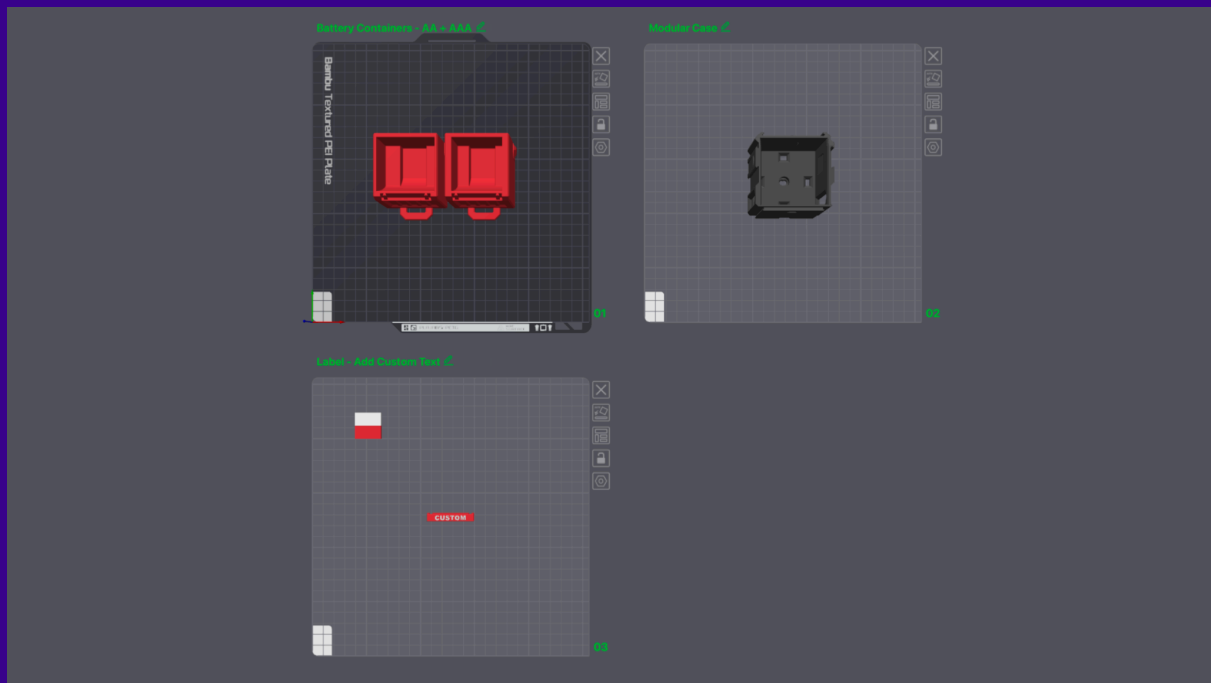
Print Settings

Important Note:

The print settings listed here are the ones that worked best for us and for our specific needs, using our own printers, materials, and environment. However, every setup is different – so you may need to adjust these settings to suit your printer, filament, and your specific needs.

Ready to Print Project

A .3mf file named **'1.Ready to Print'** is included in the download. This file is a pre-prepared Bambu Studio project, which includes the settings we used.



Individual Part Settings

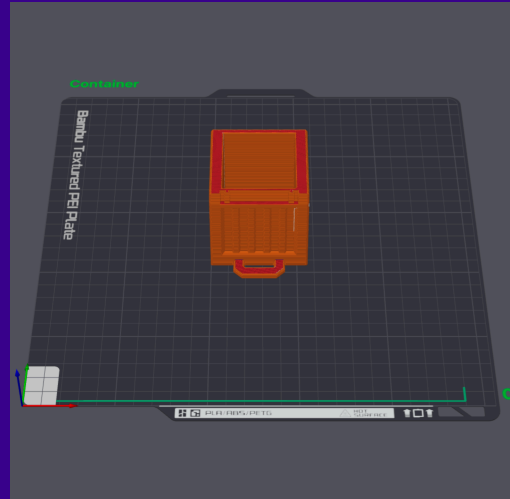
Container (AA + AAA)

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto

Estimations:

- **Print Time** - 1hr 26m
- **Filament Used** - 54g



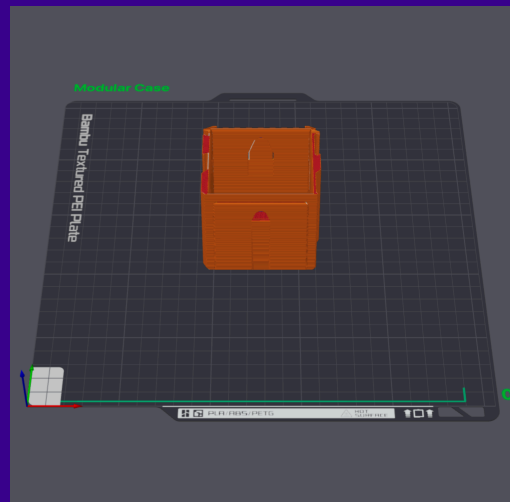
Modular Case

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto

Estimations:

- **Print Time** - 1hr 51m
- **Filament Used** - 52.71g



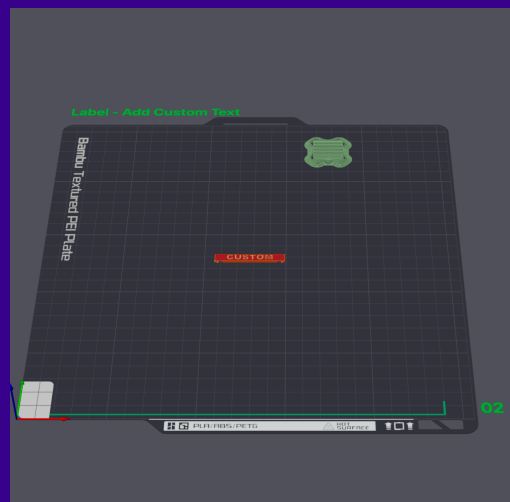
Labels (To make 'Custom Labels', see step 5)

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto

Estimations:

- **Print Time** - 5m 41s
- **Filament Used** - 2.11g



Assembly Guide

Important Note:

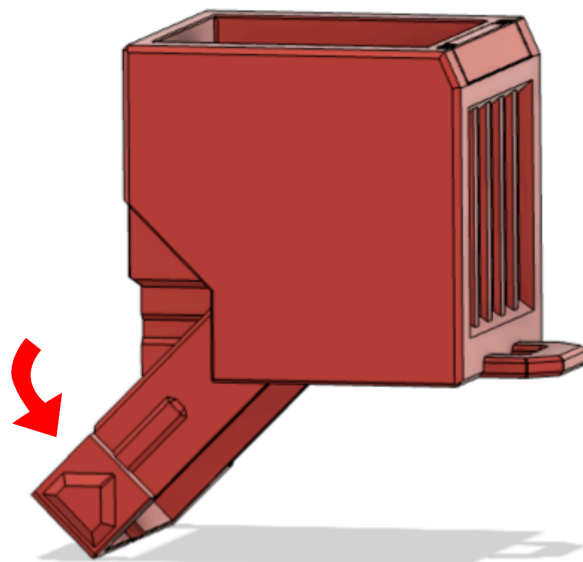
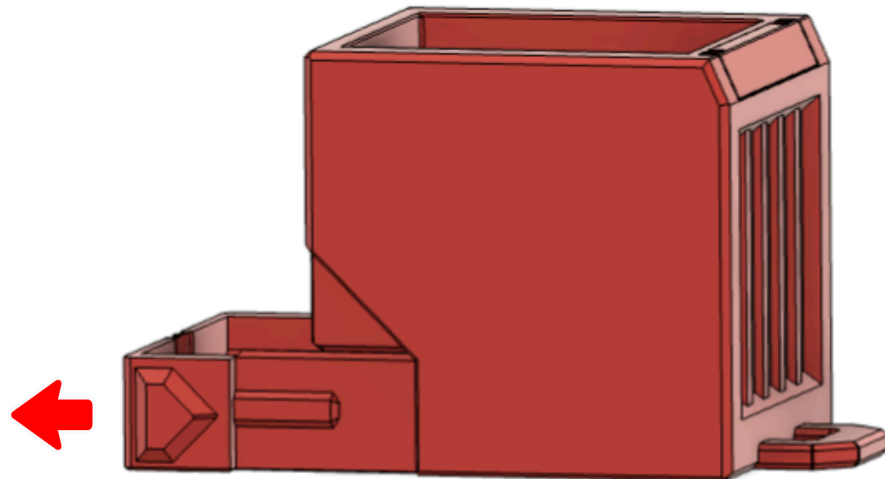
Once all parts are printed, remove any supports and check for any imperfections or poorly printed areas - *these may need post processing work or reprinting.*

Step 1 - Preparation

There are two Battery Storage Containers to choose from:

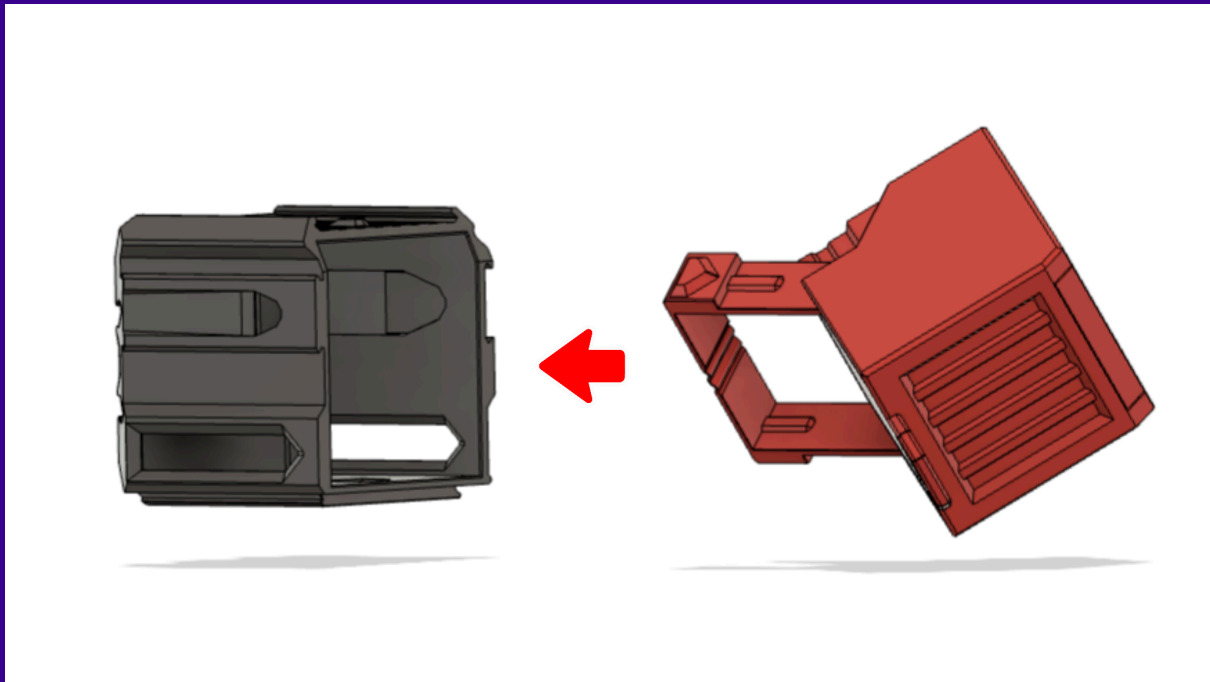
- **Container - AA Batteries**
- **Container - AAA Batteries**

Once you have printed your chosen sizes, take each container and pull the sliding hinge - it should pull free easily and slide back and forth. If it doesn't or it's stuck, check there hasn't been warping off the print bed or poor print quality.

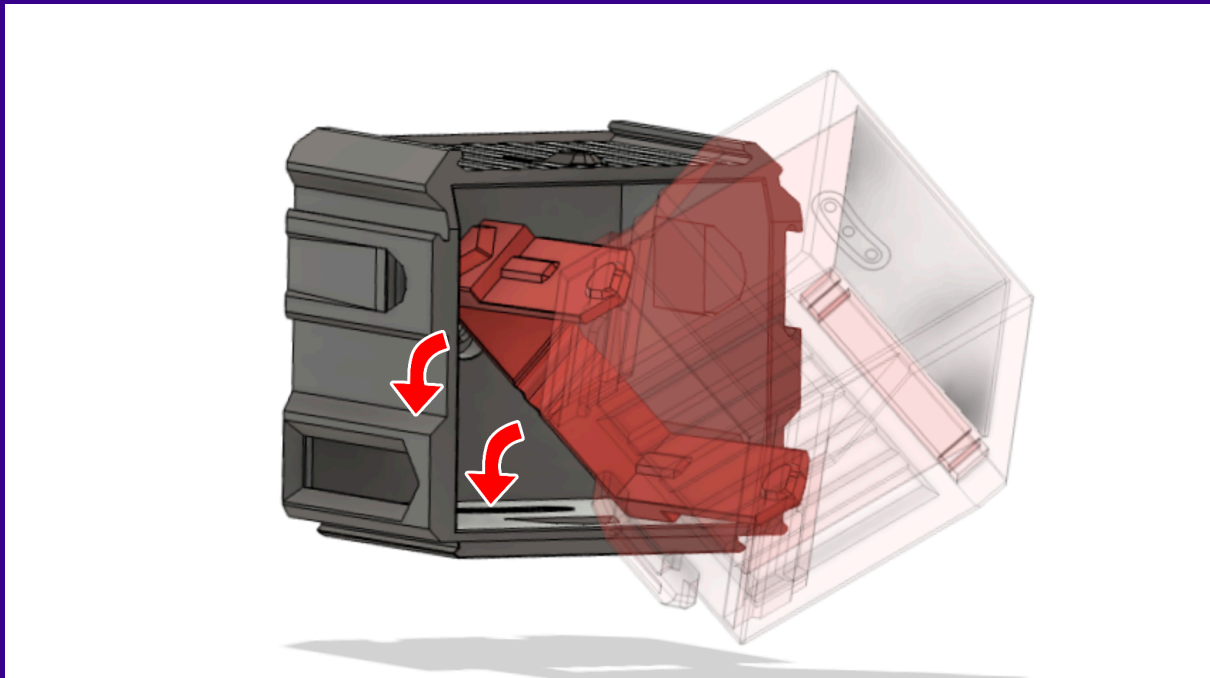


Step 2 - Assembly

Take the Modular Case and put the Container hinge in **at an angle** so it's inside the Modular Case.



Carefully turn the Container into position so that the hinge clicks into the slots either side of the Modular Case. *(We have designed a springy part into the hinge to allow it to flex slightly as it's inserted into the Modular Case).*

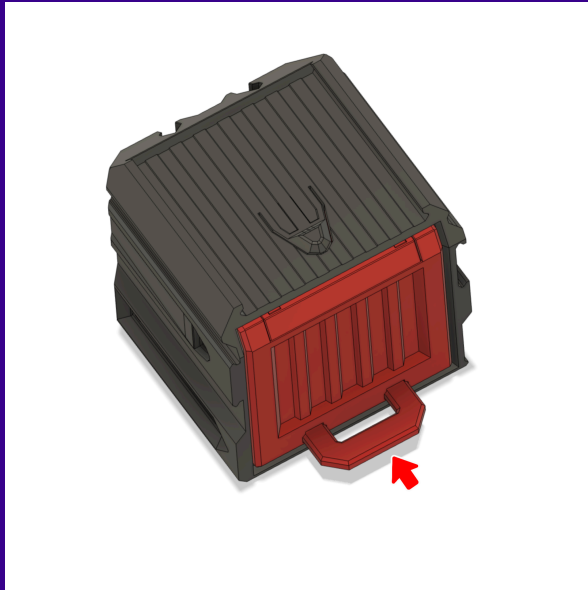


Check the Container is correctly inserted into the Modular Case by sliding it back and forth. The Container should slide freely in and out and hold at a slight angle when open.

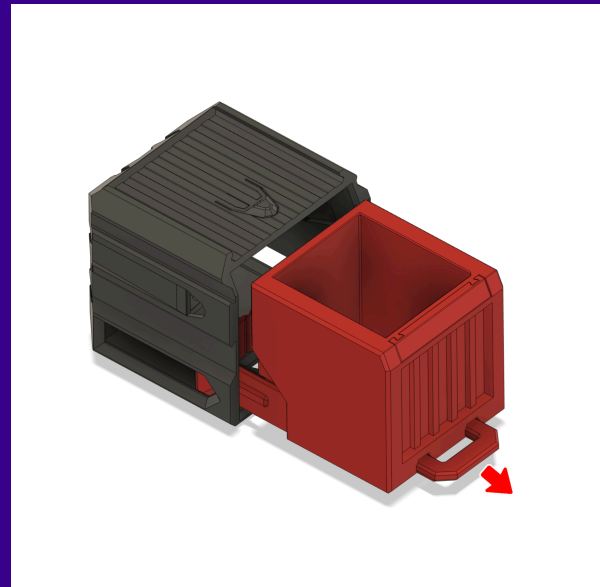
To remove the Container from the Modular Case, reverse this step, squeezing the hinge slightly inwards from either side when removing.

Step 3 - Locking Mechanism

The Container has a built-in locking mechanism in the handle. Once the container is fully closed, simply push the handle inwards and this will lock the Container into the Modular Case.



To unlock the Container, open it using the handle and you should feel a click as the lock is automatically disengaged. The handle should be fully clicked out when the container is open as shown below. To lock, simply close the container and click the handle closed as above.



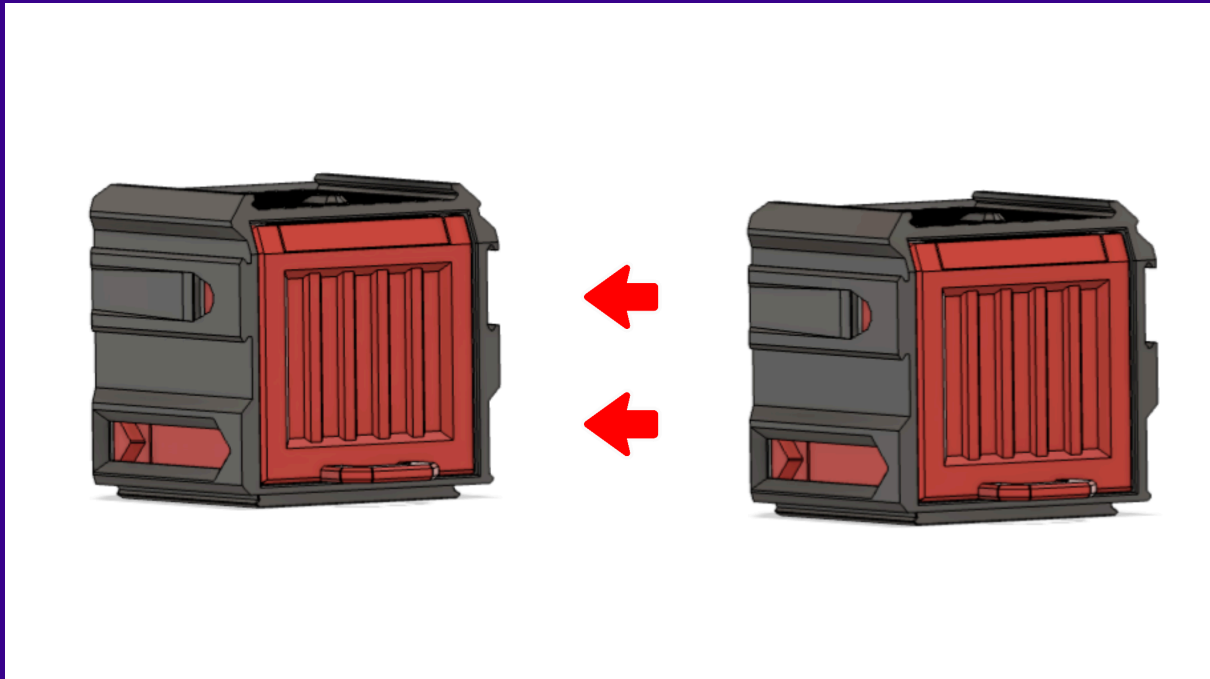
Important Note:

The locking mechanism is designed to stop the containers slightly jumping forwards when being slid around the Cargo Connect: Rails. Due to the nature of 3D printing, the lock has limited strength and should only be used for its intended purpose. For example, it is not designed to keep the Containers closed if they are physically moved around or removed from the wall system.

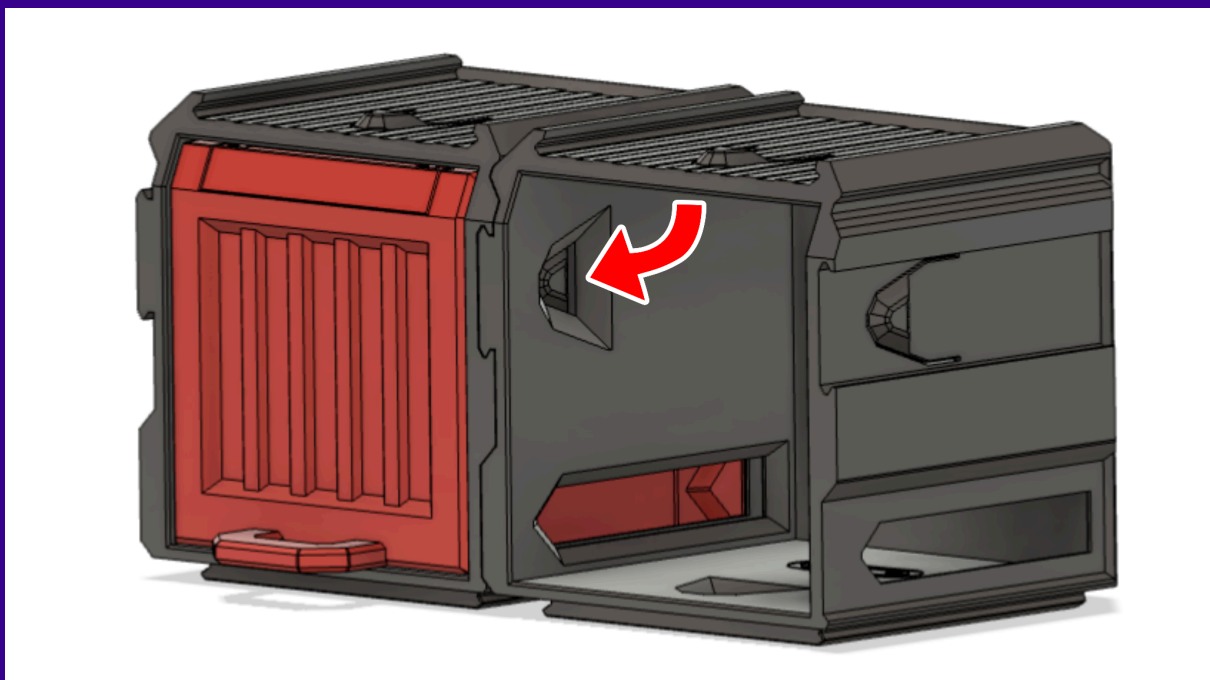
Step 4 - Modularity

Once you have your Modular Case and Container assembled, you can simply slide additional Modular Cases together (*each side will slide in one direction only due to built in stoppers, so it's best to connect the cases from left to right*).

To do this align the slots on each side and push the Modular Cases together until you hear/feel the cases click into place.



If correctly connected, the fronts of the Modular Cases will be flush with each other and the locking tab on the inside will be in the below position.

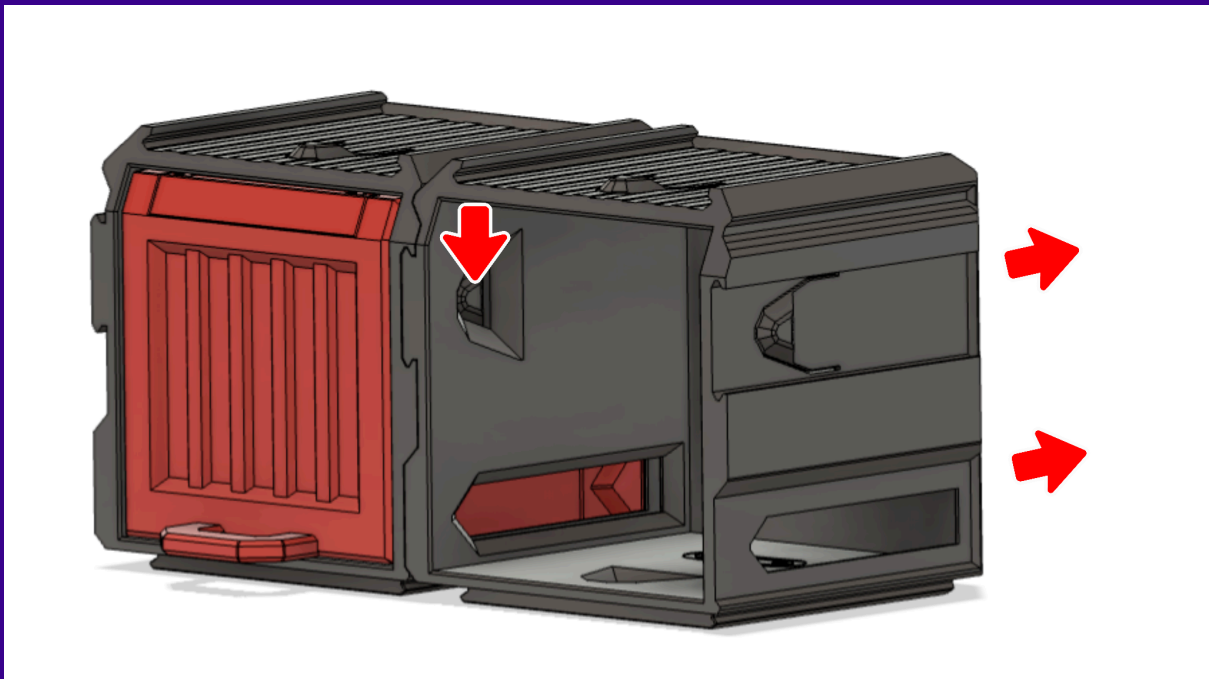


Step 4 *continued..*

The Modular Cases have a modular top, base and sides so they can be slid together from all sides to make any arrangement you desire.



To remove an Modular Case once it has been connected to another, remove the Container by reversing 'Step 2', then push the locking tab inside the Modular Case inwards whilst sliding the two Modular Cases apart.

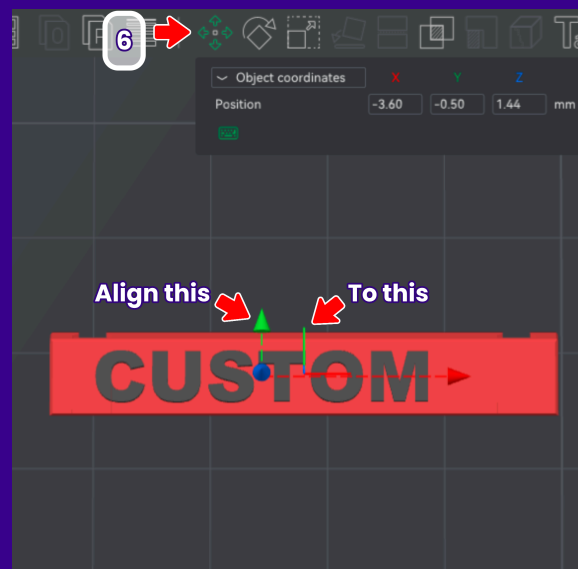
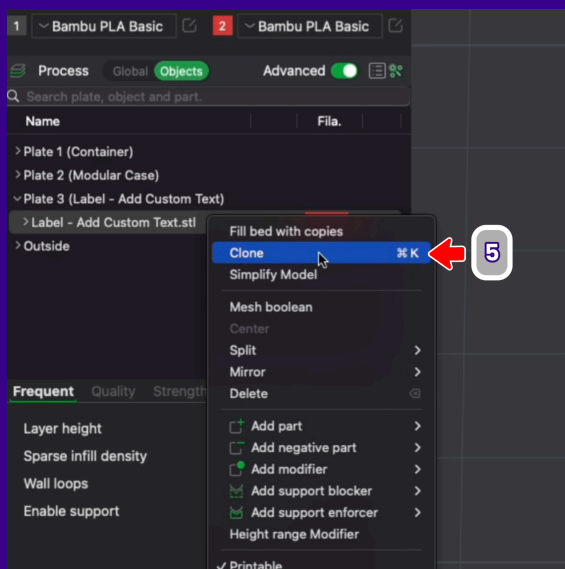
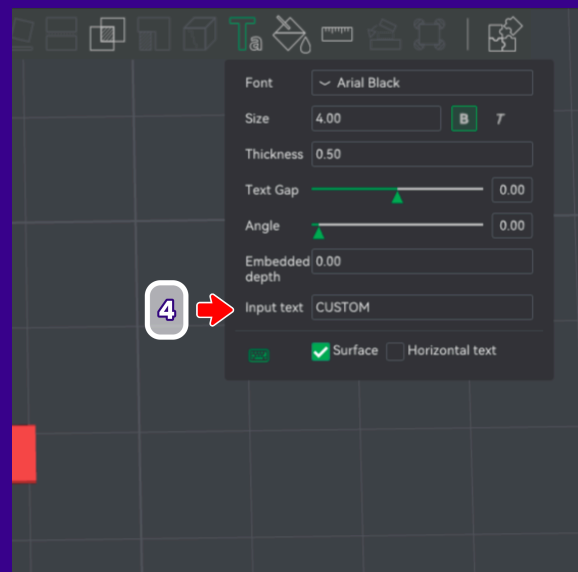
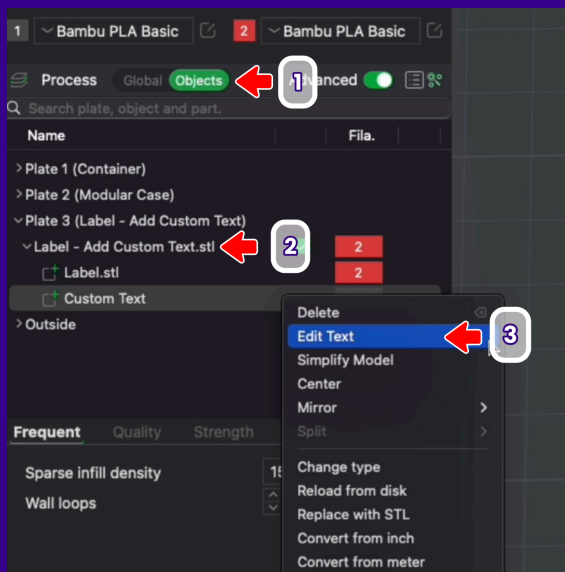


Step 5 - Custom Labels

There is a label slot on the top of each Container. We have included a preset editable label inside the '1. Ready to Print' Bambu Studio .3mf file. To alter the text in Bambu Studio, follow the steps below:

(There is also a 'Blank Label.stl' included for use with other slicers).

1. Navigate to the 'Objects' tab
2. Click on the drop down for the 'Label - Add Custom Text' STL file
3. Right click on the 'Custom Text' object and select 'Edit Text'
4. In the 'Text Box', you can change the text using the 'Input Text' box. (You can also edit the size, font etc during this process)
5. If you want to make numerous labels, you can right click on the whole 'Label - Add Custom Text' STL and select clone, then choose how many copies you need and follow the process above for each label.
6. The Custom Label object has already been centered, however when you alter the text, it will likely un-center the text horizontally. If you select the 'Custom Text' part and select 'move', you can realign the green lines shown in the picture and this should re-center the text.

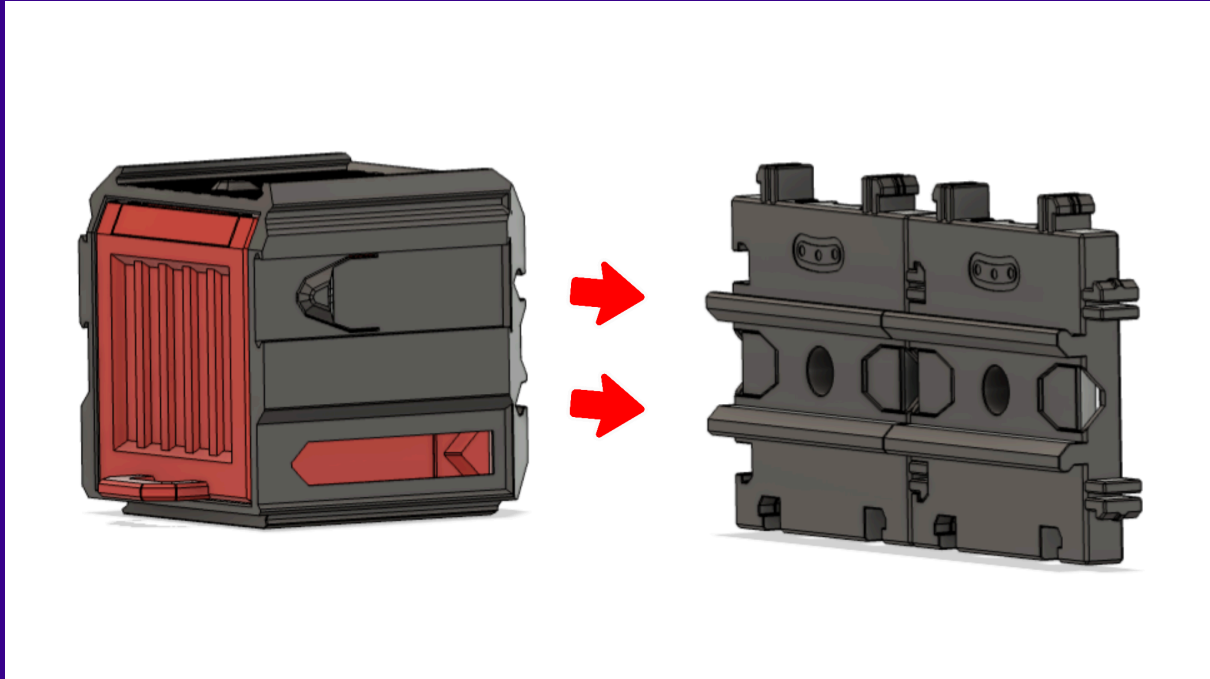


Step 6 - Mounting

Cargo Connect: Rails

The Modular Case has built-in tracks on the back that are compatible with the **Cargo Connect: Rails**. The cases simply slide on and off the Cargo Connect: Rails.

If you are connecting multiple cases together, you will need to do this **before** sliding them onto the Cargo Connect: Rails.



Step 6 *continued..*

Wall Mounting

There is a screw hole at the back of each Modular Case, should you wish to mount the Cargo Containers without using the Cargo Connect: Rails system.

On the Cargo Container Mini - 1x1, there is only one screw hole per Modular Case, so you will need to mount at least two connected cases to avoid them spinning.



Important Note on Use

The number and placement of tiles and screws used for the Cargo Modular System is **entirely at your discretion**, subject to the strength of your wall and the intended weight of your stored items.

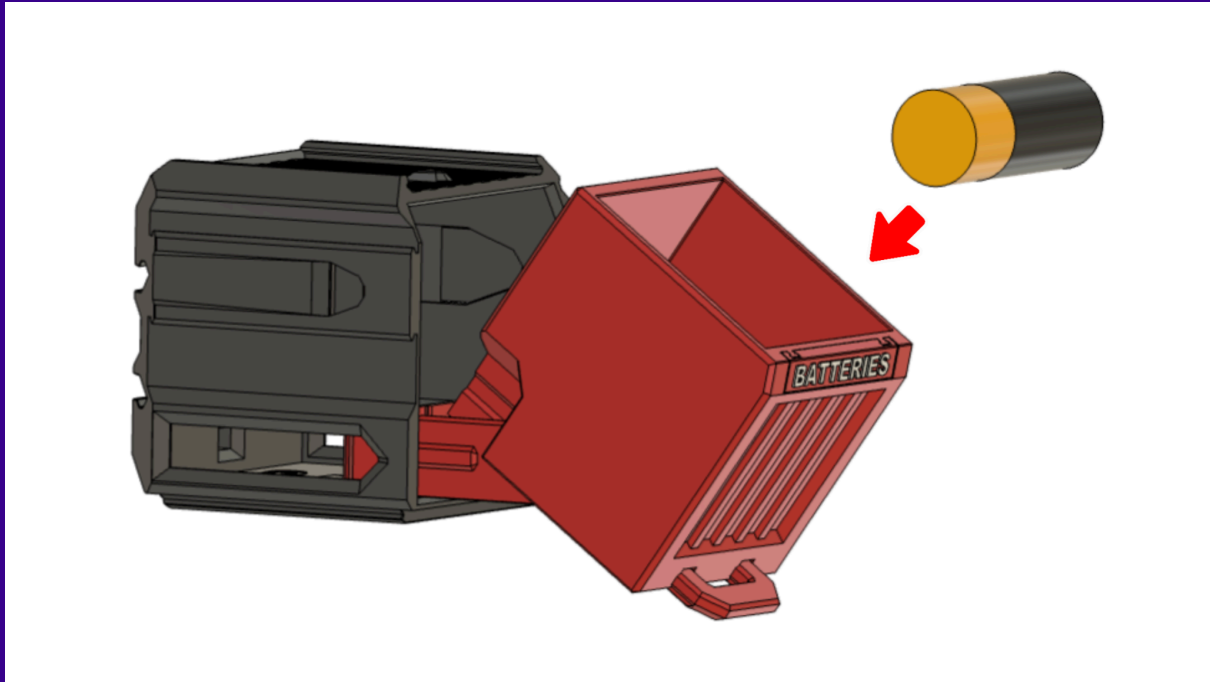
You should thoroughly test the stability and strength of your wall and the system before use and inspect the system at regular intervals to ensure the materials used have not deteriorated.

Step 7 - Use

There are two Battery Storage Containers included in the downloaded files:

- **Container – AA Batteries** – stores up to 6 AA batteries
- **Container – AAA Batteries** – stores up to 10 AAA batteries

Once printed and assembled, load the container by simply sliding AA or AAA batteries in **side-on**, as shown below. Batteries can be removed just as easily by sliding them back out.



Important Note on Battery Storage

These containers are designed purely for organizing AA and AAA batteries for easy access. They are intended to be fully 3D printed in plastic and do not provide any additional protection beyond basic organization.

Always follow the battery manufacturer's safety and storage instructions when using this product.

Troubleshooting

We put a lot of time into making our designs as easy to print and use as possible, testing extensively with various filaments and printers. That said, we understand that occasional issues can occur – below are some potential solutions that may be useful:

Warping and Bed Adhesion

Warping is a common challenge in FDM 3D printing, often caused by poor bed adhesion, rapid cooling, or inconsistent ambient temperatures.

It's important that tiles and components come off the bed as flat as possible, as warping may impact features like interlocking pegs, sliding fits, or hinge alignment. If you notice lifting corners or uneven bases, try adding a brim or applying a suitable adhesive directly to the build plate to improve bed adhesion.

Ensuring consistent ambient temperatures and proper first-layer calibration can also go a long way in reducing warping across prints.

Tolerances

The Cargo Modular System is designed to be as print-friendly as possible, and we've worked hard to make tolerances as forgiving as possible across a wide range of printers. However, the unique features and functionality of our designs can rely on tight tolerances to work as intended – especially for parts that slide, hinge, or interlock.

If you're experiencing issues with fit or movement, it's worth checking your printer calibration – including flow rate, retraction settings, and nozzle temperature – to ensure accurate extrusion. Also make sure your filament is dry and of consistent quality, and that your slicer profile is correctly set up for your printer and material.

If problems still persist, we recommend printing a tolerance test model to help identify where your setup might need fine-tuning. These are widely available from reputable sources such as Thangs.

Use Disclaimer

Every Play Conveyor model has been designed and tested with care, but results will vary depending on your printer, materials, settings, and environment. Because of this, we cannot guarantee print outcomes, weight capacities, or the structural performance of any printed items.

You are responsible for ensuring that both the printing and assembly of any models are completed to a safe and functional standard. Always test your prints — especially components that are wall-mounted, load-bearing, or move — and check them regularly for wear or failure.

This user guide is provided as a general reference to support your build, but it is not exhaustive or guaranteed to suit all applications. Use your judgment where needed to suit your specific setup.

All files are provided “as is” and without warranty. Play Conveyor Ltd accepts no responsibility for any injury, damage, or loss resulting from the use or misuse of our designs.

This disclaimer highlights some of the key points when using our models, but it does not replace or override the official ‘License Agreement’.

The ‘License Agreement’ associated with your subscription tier is the official document that defines your permitted use of the model, any restrictions, and your responsibilities. Please ensure you review the relevant license agreement to confirm proper usage.

Thangs Members

- Maker tier: covered by the **Personal License** (for personal use only)
- Merchant tier: covered by the **Commercial License** (includes permitted commercial use)

Your applicable license can be found under each model’s listing on our Thangs page.

Patreon Members

- The license for all Patreon ‘Maker Access’ tiers is included within the downloadable files for each design.

It is your responsibility to review and follow the terms of your applicable license.